

Rules for removing brackets, and the coefficient

A plus leaves the signs alone; a minus changes every one of them.

LESSON 59

1 × 40 MIN

MATEMATIKA 7, §25

S8 2.2

LESSON CLOCK

10'

12'

12'

6'

Plus and minus before a bracket	10 min
A number before a bracket	12 min
The coefficient	12 min
Homework	6 min

LEARNING OBJECTIVES

- Remove brackets preceded by a plus and by a minus sign.
- Multiply a bracket by a number using the distributive law.
- Identify and use the coefficient of a term.
- Simplify an expression by removing brackets and collecting terms.

TEXTBOOK

National	pp. 156–159
Cambridge	Stage 8, pp. 16–20
Workbook	Exercise 2.2

01

Explanation

Plus and minus before a bracket

$$a + (b - c) = a + b - c$$

$$a - (b - c) = a - b + c$$

EXPRESSION	BRACKETS REMOVED
$5 + (x - 3)$	$5 + x - 3$
$5 - (x - 3)$	$5 - x + 3$
$-(2a - b + 4)$	$-2a + b - 4$
$7 - (-x + 2)$	$7 + x - 2$

A MINUS CHANGES EVERY SIGN INSIDE

$-(2a - b + 4) = -2a + b - 4$ — all three. Changing only the first is the error this lesson exists to prevent, and it recurs all year.

A number before a bracket

$$k(a + b) = ka + kb$$

Every term inside is multiplied — including the last one, and including the signs.

EXPRESSION	EXPANDED
$3(x + 4)$	$3x + 12$
$3(x - 4)$	$3x - 12$
$-3(x - 4)$	$-3x + 12$
$-2(3a - b + 5)$	$-6a + 2b - 10$

$- (...)$ IS $-1(...)$

The two rules of this lesson are one rule: a minus in front is multiplication by -1 , and the distributive law does the rest.

The coefficient

The **coefficient** is the numerical factor of a term, taken with its sign.

TERM	COEFFICIENT	LETTERS
$7ab$	7	ab
$-3x^2y$	-3	x^2y
a	1	a
$-b$	-1	b
$\frac{2x}{3}$	$\frac{2}{3}$	x

Terms with the **same letters to the same powers** are **like terms**, and only like terms can be collected: $3x + 5x = 8x$, but $3x + 5y$ cannot be simplified.

$3x$ AND $3x^2$ ARE NOT LIKE TERMS

The powers must match too. Adding them gives $3x + 3x^2$, which is already as simple as it gets.

02 Worked examples

EXAMPLE 1 Remove the brackets: $5 - (x - 3)$ and $-(2a - b + 4)$.

1 A minus before the bracket changes every sign.

2 $5 - x + 3 = 8 - x$

3 $-2a + b - 4$
All three signs.

4 Nothing further to collect in the second.

ANSWER

$8 - x$ and $-2a + b - 4$

EXAMPLE 2 Expand $-2(3a - b + 5)$.

1 Multiply every term by -2 .

2 $-2 \times 3a = -6a$

3 $-2 \times (-b) = +2b$

4 $-2 \times 5 = -10$, giving $-6a + 2b - 10$.

ANSWER

$-6a + 2b - 10$

EXAMPLE 3 Simplify $4(x + 2) - 3(x - 1)$.

$$① \quad 4x + 8$$

$$② \quad -3x + 3$$

Both signs change.

$$③ \quad 4x + 8 - 3x + 3$$

$$④ \quad = x + 11$$

ANSWER

$$x + 11$$

03 Interactive model

Write $-(a - b)$ on the board and ask for its value at $a = 5$, $b = 2$ before removing the brackets; the numerical check settles the sign rule.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Bracket	Qavs	Скобка
To remove brackets	Qavslarni ochish	Раскрыть скобки
Distributive law	Taqsimot qonuni	Распределительный закон
Coefficient	Koeffitsiyent	Коэффициент
Like terms	O'xshash hadlar	Подобные члены
To simplify	Soddalashtirish	Упростить
Sign change	Ishora o'zgarishi	Изменение знака
Nested brackets	Ichma-ich qavslar	Вложенные скобки

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$5 - (x - 3)$ equals:

$-(2a - b + 4)$ equals:

$-3(x - 4)$ equals:

The coefficient of $-3x^2y$:

3

-3

x^2y

1

The coefficient of $-b$:

0

1

-1

b

Are $3x$ and $3x^2$ like terms?

yes

no

sometimes

only if $x = 1$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $5 + (x - 3)$

ANSWER $x + 2$

2. $5 - (x - 3)$

ANSWER $8 - x$

3. $3(x + 4)$

ANSWER $3x + 12$

4. $3(x - 4)$

ANSWER $3x - 12$

5. $-3(x - 4)$

ANSWER $-3x + 12$

6. Coefficient of $7ab$

ANSWER 7

7. Coefficient of $-b$

ANSWER -1 **Medium · the standard question**

7 PROBLEMS

1. $-(2a - b + 4)$

ANSWER $-2a + b - 4$

2. $-2(3a - b + 5)$

ANSWER $-6a + 2b - 10$

3. $7 - (-x + 2)$

ANSWER $x + 5$

4. $4(x + 2) - 3(x - 1)$

ANSWER $x + 11$

5. $2(a - 3) + 5(a + 1)$

ANSWER $7a - 1$

6. Collect $3x + 5x - 2x$

ANSWER $6x$

7. Collect $3x + 5y$

ANSWER Already simplest

Hard · stretch and reason

7 PROBLEMS

1. $3(2x - 1) - 2(3x - 4)$

ANSWER 5

2. $-[2 - (3 - x)]$

ANSWER $1 - x$

3. $5 - 2(x - 3(x + 1))$

ANSWER $4x + 11$

4. $a - (b - (c - d))$

ANSWER $a - b + c - d$

5. Simplify $4(x + y) - 4(x - y)$

ANSWER $8y$

6. Evaluate $-(a - b)$ at $a = 5, b = 2$

ANSWER -3

7. Coefficient of $\frac{3x}{4}$

ANSWER $\frac{3}{4}$ **07 Homework****Homework — 5 tasks**

Check every expansion by substituting one easy pair of values.

1. Remove the brackets: $8 - (y - 5)$ and $-(3m - 2n + 1)$.
2. Expand $-4(2a - 3b + 1)$.
3. Simplify $5(x - 2) - 2(x + 3)$.
4. Give the coefficients in $-5a + b - \frac{c}{2}$.

5. Simplify $3(a + b) - 3(a - b)$.